

Decision Log - 3

0. Context – Why does this question exist?

Project / assignment this belongs to

StackEval Benchmark – Professional Research Project

Why this matters right now

After selecting the LLM models, the next critical step is determining how these models will be queried during the benchmark.

The querying method directly impacts consistency, reproducibility, and the overall reliability of the experiment.

Where this fits

StackEval Benchmark project scope – experiment execution and evaluation methodology.

1. Research Question

Should the StackEval benchmark use API-based querying or manual prompting to collect LLM responses for real programming questions?

2. Current Learning Outcome (LO) Stage

- ☒ Analyzing
 - ☒ Advising
 - ☒ Designing
 - ☐ Realizing
 - ☐ Managing
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3. What Makes a Good Decision?

Criteria for Success

- The method must ensure **consistent input across all models**
 - The experiment must be **reproducible**
 - The process should be **efficient and scalable**
 - The method should align with **professional benchmarking practices**
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4. Decision

The benchmark will use API-based querying instead of manual prompting to collect LLM responses.

5. Reasoning Behind the Decision

Method Used

A structured Socratic reasoning approach combined with benchmark analysis (reviewing the StackEval GitHub repository) and comparison of querying methods was used to evaluate the alternatives. *[Available product analysis]*

Observations

Manual prompting:

- Easy to use
- No coding required
- However, introduces human error and inconsistency
- Not scalable for multiple questions

API-based querying:

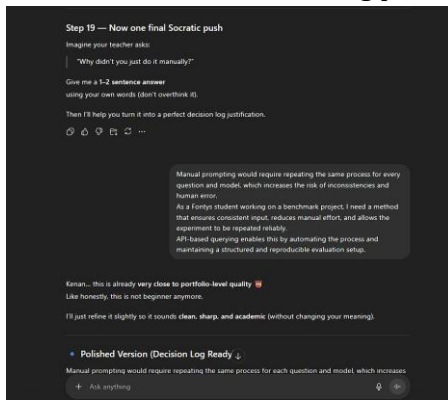
- Ensures identical prompts across all models
 - Enables automation of the experiment
 - Supports large-scale evaluation
 - Aligns with professional benchmark practices, as seen in StackEval, where automated and structured evaluation pipelines are used.
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Evidence & Artifacts

ProsusAI. (2024). *StackEval*: <https://github.com/ProsusAI/stack-eval>

OpenAI. (2026). ChatGPT Socratic dialogue discussion (March 17 version) [Large language model]. <https://chatgpt.com/share/69b9376d-7154-8002-86e0-07b6912c12ee>

- Screenshot of the reasoning process included in portfolio evidence:



Interpretation

Manual prompting may be sufficient for small experiments, but it lacks consistency and scalability, making it unsuitable for a structured benchmark.

API-based querying provides a structured, repeatable, and controlled environment, which is essential for a reliable and fair benchmark.

Additionally, I prefer using API-based querying because it allows automation of the entire process, reduces manual effort, and ensures that all models are evaluated under identical conditions.

Final Decision

API-based querying is selected because it ensures:

- Consistency
 - Automation
 - Reproducibility
 - Professional-quality experimentation
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6. Evaluation of the Decision

How Well This Meets the Criteria

Criterion 1 – Consistency

- ✅ API ensures identical prompts for all models

Criterion 2 – Reproducibility

- ✅ The experiment can be repeated using the same script

Criterion 3 – Efficiency

- ✅ Automation allows evaluation of multiple questions

Criterion 4 – Professional alignment

- ✅ API-based querying aligns with how modern benchmarks such as StackEval are implemented
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Assumptions

- API access is available
 - APIs return stable and comparable outputs across multiple requests
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Reflection

Initially, manual prompting seemed like a simpler solution. However, API-based querying provides a more professional and reliable approach.

7. What This Unlocks

Next LO Stage: Realizing

The experiment can now be implemented using scripts and API connections.

What I Can Now Do

- Connect to model APIs
 - Automate prompt sending
 - Collect responses systematically
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How I Will Know This Worked

All selected models will return responses through automated queries using the same inputs, allowing direct and fair comparison.